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Research on Coordinated Active and Reactive Power Control of VSC-HVDC Connected Renewable Energy System for Enhancing Static Voltage Stability

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***Abstract*—The static voltage stability studies of the AC system with VSC-HVDC (Voltage Source Converter-based High Voltage Direct Current) connected renewable energy mainly focus on reactive power control. This paper proposes a coordinated active and reactive power control strategy based on the L-index to improve the static voltage stability of the receiving-end AC system. Firstly, for a power system including the VSC-HVDC connected renewable energy, the L-index and its sensitivity to different variables are calculated. Secondly, an optimization model for coordinated control of the VSC-HVDC connected renewable energy is established by using the L-index of the receiving-end system as the optimization objective, and power flow, node voltage, and transmission power as constraints. Finally, the optimization model is solved using the Particle Swarm Optimization (PSO) algorithm to obtain the optimal active and reactive power support values. Simulation results show that the coordinated control method can significantly improve the static voltage stability of the receiving-end AC system while ensuring the safe operation of the system.**

Keywords—VSC-HVDC connected Renewable Energy, Coordinated Active and Reactive Power Control, Static Voltage Stability, L Index

I. INTRODUCTION

With the implementation of the "dual carbon" goals, the proportion of renewable energy has increased significantly. Flexible DC transmission technology, with its advantages in integrating renewable energy, has become an important component of modern power systems. However, this also poses greater challenges to system voltage stability[1-3]. In normal working conditions, the onshore converter station lacks active response and support for grid voltage disturbances, and the dynamic response capabilities for active and reactive power of the flexible DC system cannot be fully utilized[4]. Therefore, it is essential to propose a coordinated active and reactive power control strategy for the power system including the VSC-HVDC connected renewable energy .

Static voltage stability is an important indicator for measuring the ability of a power system to maintain stable voltage levels under normal operating conditions. The main research methods for static voltage stability include the continuation power flow method[5], sensitivity analysis[6], eigenvalue analysis[7], singular value analysis[8], L index, and artificial intelligence methods[9]. Among these methods, the L index is widely used due to its fast computation speed, clear physical meaning, and sensitivity to parameter changes. Reference[10] combined power flow equations to propose the sensitivity of the L index to load and the relationship between the L index and eigenvalues, and based on this, pointed out methods to determine the threshold of the L index. Reference[11] proposed an improved PEM to calculate the moments and semi-invariants of the L index for calculating voltage instability risk. Reference[12] constructed a static voltage stability assessment model based on the L index using extended affine arithmetic, and then quantitatively evaluated the importance of each variable's impact on static voltage stability. Although the above references have reasonably assessed the static voltage stability performance based on the L index, they are all limited to AC systems and rarely consider the impact of renewable energy integration via VSC-HVDC.

Currently, research on the static voltage stability of a power system including the VSC-HVDC connected renewable energy mostly focuses on reactive power. Reference[13] uses the Jacobian matrix at the system voltage collapse point to derive the sensitivity of the system load margin to VSC-HVDC control parameters. It prioritizes adjusting the parameters with higher sensitivity to enhance static voltage stability. Reference[14] identifies critical nodes and lines in AC/DC hybrid systems through modal analysis and improves the overall system's static voltage stability by providing reactive power compensation to

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these nodes or lines. Reference[15] establishes an optimization model with voltage deviation, L index, active power loss, and total reactive power compensation as optimization objectives to solve the reactive power compensation capacity for offshore wind power integration via VSC-HVDC systems. It can be seen that the primary focus of all the above references is given to the optimization of reactive power compensation, neglecting the contribution from active power regulation of the converter stations, which will be shown in the paper to play a significant role in system static voltage stability as well.

In light of the shortcomings of previous research, this paper proposes a coordinated active and reactive power control strategy based on the L index. Firstly, for a power system including the VSC-HVDC connected renewable energy, the L index and its sensitivity to different variables are calculated based on the power flow equations, and the renewable energy integration via VSC-HVDC is modeled as a power source. Then, an optimization model for coordinated control of the VSC-HVDC connected renewable energy is established. The optimization model is solved using the PSO algorithm to obtain the optimal active and reactive power support capacities. Finally, the effectiveness of the proposed coordinated control method are verified in a model containing a VSC-HVDC system.

II. L INDEX AND CORRESPONDING SENSITIVITY

A. Static Voltage Stability Analysis Based on L Index

The node voltage equations of the power network are established based on Kirchhoff's Current Law：

$$\begin{bmatrix} \boldsymbol{I}_G \\ \boldsymbol{I}_L \\ \boldsymbol{0} \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y}'_{GG} & \boldsymbol{Y}'_{GL} & \boldsymbol{Y}'_{GN} \\ \boldsymbol{Y}'_{LG} & \boldsymbol{Y}'_{LL} & \boldsymbol{Y}'_{LN} \\ \boldsymbol{Y}'_{NG} & \boldsymbol{Y}'_{NL} & \boldsymbol{Y}'_{NN} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_G \\ \boldsymbol{U}_L \\ \boldsymbol{U}_N \end{bmatrix} \tag{1}$$

where $\boldsymbol{I}_G$ and $\boldsymbol{I}_L$ are the current vectors of the generator nodes and load nodes, respectively; $\boldsymbol{U}_G$, $\boldsymbol{U}_L$, and $\boldsymbol{U}_N$ are the voltage vectors of the generator nodes, load nodes, and interconnection nodes, respectively; $\boldsymbol{Y}'_{GG}, \boldsymbol{Y}'_{GL}, \boldsymbol{Y}'_{GN}, \boldsymbol{Y}'_{LG}, \boldsymbol{Y}'_{LL}, \boldsymbol{Y}'_{LN}, \boldsymbol{Y}'_{NG}, \boldsymbol{Y}'_{NL}, \boldsymbol{Y}'_{NN}$ are the sub-matrices of the node admittance matrix.

By eliminating the interconnection nodes, (1) can be transformed into:

$$\begin{bmatrix} \boldsymbol{I}_G \\ \boldsymbol{I}_L \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y}_{GG} & \boldsymbol{Y}_{GL} \\ \boldsymbol{Y}_{LG} & \boldsymbol{Y}_{LL} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_G \\ \boldsymbol{U}_L \end{bmatrix} \tag{2}$$

where $\boldsymbol{Y}_{GG}, \boldsymbol{Y}_{GL}, \boldsymbol{Y}_{LG}, \boldsymbol{Y}_{LL}$ are the sub-matrix blocks of the node admittance matrix without interconnection nodes, and their calculation formulas are:

$$\begin{cases} \boldsymbol{Y}_{GG} = \boldsymbol{Y}'_{GG} - \boldsymbol{Y}'_{GN}\boldsymbol{Y}'^{-1}_{NN}\boldsymbol{Y}'_{NG} \\ \boldsymbol{Y}_{GL} = \boldsymbol{Y}'_{GL} - \boldsymbol{Y}'_{GN}\boldsymbol{Y}'^{-1}_{NN}\boldsymbol{Y}'_{NL} \\ \boldsymbol{Y}_{LG} = \boldsymbol{Y}'_{LG} - \boldsymbol{Y}'_{LN}\boldsymbol{Y}'^{-1}_{NN}\boldsymbol{Y}'_{NG} \\ \boldsymbol{Y}_{LL} = \boldsymbol{Y}'_{LL} - \boldsymbol{Y}'_{LN}\boldsymbol{Y}'^{-1}_{NN}\boldsymbol{Y}'_{NL} \end{cases} \tag{3}$$

Define the impedance matrix of the load nodes $\boldsymbol{Z}_{LL} = \boldsymbol{Y}_{LL}^{-1}$ and the participation factor matrix of the load nodes $\boldsymbol{F}_{LG} = -\boldsymbol{Y}_{LL}^{-1}\boldsymbol{Y}_{LG}$, (3) can be transformed into:

$$\begin{bmatrix} \boldsymbol{I}_G \\ \boldsymbol{U}_L \end{bmatrix} = \begin{bmatrix} \boldsymbol{Y}_{GG} - \boldsymbol{Y}_{GL}\boldsymbol{Z}_{LL}\boldsymbol{Y}_{LG} & \boldsymbol{Y}_{GL}\boldsymbol{Z}_{LL} \\ \boldsymbol{F}_{LG} & \boldsymbol{Z}_{LL} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_G \\ \boldsymbol{I}_L \end{bmatrix} \tag{4}$$

From (4), the voltage vector for each load node can be obtained, and its calculation formula is:

$$U_{\mathrm{L},j} = \sum_{k \in N_{\mathrm{L}}} Z_{jk} I_{\mathrm{L},k} + \sum_{i \in N_{\mathrm{G}}} F_{ji} U_{\mathrm{G},i} \tag{5}$$

where $U_{L,j}$ is the voltage vector of the j-th load node; Z_{jk} is the element of matrix $\boldsymbol{Z}_{LL}$; F_{ji} is the element of the matrix $\boldsymbol{F}_{LG}$; $I_{L,k}$ is the current vector of the k-th load node; $U_{G,i}$ is the voltage vector of the i-th generator node.

The voltage vector of the load node in (5) is divided into two parts, one related to the current vector of the load node and the other related to the voltage vector of the generator node. The local voltage stability index L_j is defined as the modulus of the complex ratio of the former to $U_{L,j}$, and its calculation formula is:

$$L_j = \left| 1 - \frac{\sum_{i \in N_{\mathrm{G}}} F_{ji} U_{\mathrm{G},i}}{U_{\mathrm{L},j}} \right| = \left| \frac{\sum_{k \in N_{\mathrm{L}}} Z_{jk} I_{\mathrm{L},k}}{U_{\mathrm{L},j}} \right| \tag{6}$$

Therefore, the set of local voltage stability indices for all load nodes in the power network can be obtained, denoted as $L' = [L_1, L_2, \ldots, L_n]$. The voltage stability index for the entire power network is defined as:

$$L = \| L' \|_{\infty} \tag{7}$$

If $L<1$, the system voltage is stable; if $L>1$, the system voltage is unstable; if $L=1$, the system is critically stable.

B. Sensitivity Analysis of the L Index

The power flow equations are defined as below

$$\begin{bmatrix} \boldsymbol{\Delta P} \\ \boldsymbol{\Delta Q} \end{bmatrix} + \boldsymbol{J} \begin{bmatrix} \boldsymbol{\Delta \theta} \\ \boldsymbol{\Delta U} \end{bmatrix} = \boldsymbol{0} \tag{8}$$

The L index primarily depends on the degree of loading and the voltage level, and it is also affected by the load, generator output power, and the grid structure. Equation (6) can be further expressed as:

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$$\begin{cases} L_j = \left| L_{\mathrm{R}j} + \mathrm{j}L_{\mathrm{I}j} \right| \\ L_{\mathrm{R}j} = 1 - \dfrac{\sum\limits_{i \in N_{\mathrm{G}}} \left| F_{ji} \right| U_{\mathrm{G},i} \cos\theta_{jif}}{U_{\mathrm{L},j}}, j \in N_{\mathrm{L}} \\ L_{\mathrm{I}j} = \dfrac{\sum\limits_{i \in N_{\mathrm{G}}} \left| F_{ji} \right| U_{\mathrm{G},i} \sin\theta_{jif}}{U_{\mathrm{L},j}}, j \in N_{\mathrm{L}} \\ \theta_{jif} = \theta_j - \theta_i - \theta_f \end{cases} \tag{9}$$

where the magnitude and angle of F_{ji} are denoted as $|F_{ji}|$ and θ_f, θ_j, θ_i are the phase angles of the load node voltage and the generator node voltage. The partial derivatives of L_j with respect to L_{Rj} and L_{Ij} are:

$$\begin{cases} \dfrac{\partial L_j}{\partial L_{\mathrm{R}j}} = \dfrac{L_{\mathrm{R}j}}{\sqrt{L_{\mathrm{R}j}^2 + L_{\mathrm{I}j}^2}} \\ \dfrac{\partial L_j}{\partial L_{\mathrm{I}j}} = \dfrac{L_{\mathrm{I}j}}{\sqrt{L_{\mathrm{R}j}^2 + L_{\mathrm{I}j}^2}} \end{cases} \tag{10}$$

By linearly expanding L_{Rj} and L_{Ij} with respect to θ_j, θ_i and U_j, (11) can be obtained.

$$\begin{bmatrix} \boldsymbol{\Delta L_{Rj}} \\ \boldsymbol{\Delta L_{Ij}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{T_{Rj,\theta_i}} & \boldsymbol{T_{Rj,\theta_j}} & \boldsymbol{T_{Rj,U_j}} \\ \boldsymbol{T_{Ij,\theta_i}} & \boldsymbol{T_{Ij,\theta_j}} & \boldsymbol{T_{Ij,U_j}} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Delta\theta_i} \\ \boldsymbol{\Delta\theta_j} \\ \boldsymbol{\Delta U_j} \end{bmatrix} = \boldsymbol{T} \begin{bmatrix} \boldsymbol{\Delta\theta} \\ \boldsymbol{\Delta U} \end{bmatrix} \tag{11}$$

where U_i and U_j are the magnitudes of the voltage vectors at the generator nodes and load nodes, respectively.

By combining (8) with (11), we can obtain:

$$\begin{bmatrix} \boldsymbol{\Delta L_{Rj}} \\ \boldsymbol{\Delta L_{Ij}} \end{bmatrix} = \boldsymbol{-TJ^{-1}} \begin{bmatrix} \boldsymbol{\Delta P} \\ \boldsymbol{\Delta Q} \end{bmatrix} \tag{12}$$

By combining (10) with (12), (13) can be obtained, which represents the sensitivity of the L index at the load point to the active and reactive power at each node m.

$$\begin{cases} \dfrac{\partial L_j}{\partial P_m} = \dfrac{\partial L_j}{\partial L_{\mathrm{I}j}} \dfrac{\partial L_{\mathrm{I}j}}{\partial P_m} + \dfrac{\partial L_j}{\partial L_{\mathrm{R}j}} \dfrac{\partial L_{\mathrm{R}j}}{\partial P_m} \\ \dfrac{\partial L_j}{\partial Q_m} = \dfrac{\partial L_j}{\partial L_{\mathrm{I}j}} \dfrac{\partial L_{\mathrm{I}j}}{\partial Q_m} + \dfrac{\partial L_j}{\partial L_{\mathrm{R}j}} \dfrac{\partial L_{\mathrm{R}j}}{\partial Q_m} \end{cases} \tag{13}$$

Based on the above steps, the sensitivity index of the L index to each control variable can be calculated.

III. Coordinated Active and Reactive Power Control for VSC-HVDC Connected Renewable Energy to Enhance Static Voltage Stability

A. Equivalence of VSC-HVDC Connected Renewable Energy

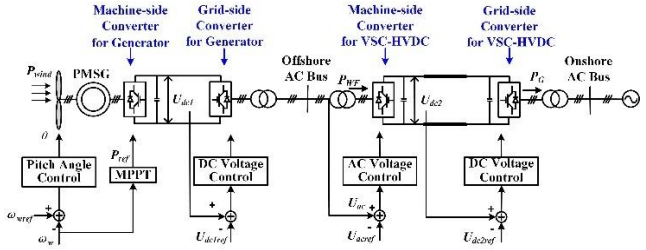


Fig. 1 The Grid-Connection Topology of Offshore Wind Power via VSC-HVDC Systems

Taking the wind power integrated through VSC-HVDC as an example, the machine-side converter for wind turbine typically employs Maximum Power Point Tracking (MPPT) control to maximize the capture of wind energy. The grid-side converter for wind turbine generally uses constant DC voltage control to maintain the DC bus at a set reference value. The machine-side converter for VSC-HVDC system often adopts constant AC voltage control to maintain the AC bus voltage of the wind farm at a set value. This provides stable voltage support for the wind turbines. The control method of grid-side converter for VSC-HVDC is similar to grid-side converter for generator.

Under the aforementioned control and neglecting system losses, the active power output P_G of the VSC-HVDC system is determined by the P_{ref} on the wind turbine side, while the reactive power output $Q_{vsc,r}$ of the VSC-HVDC system is closely related to the control mode of the converter station. Typical reactive power control modes include constant reactive power control and constant AC voltage control.

When the reactive power control mode is set to constant reactive power control, the specific expression for $Q_{vsc,r}$ is as follows:

$$Q_{vsc,r} = Q_{ref} \tag{14}$$

where Q_{ref} is the reference value of the reactive power of the converter station.

When the reactive power control mode is set to constant AC voltage control, the specific expression for $Q_{vsc,r}$ is as follows:

$$Q_{vsc,r} = \frac{U_{ac,ref}^2}{X_{ac,eq}} \tag{15}$$

where $U_{ac,ref}$ is the reference value of the AC voltage of the converter station, and $X_{ac,eq}$ is the equivalent reactance of the AC system.

Therefore, this paper models the interconnection point of VSC-HVDC connected renewable energy in the receiving-end AC system as a power source.

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B. Optimization Model for Coordinated Power Control

In order to balance the transmission efficiency of the VSC-HVDC system and the static voltage stability of the AC system, an optimization model is constructed in the following.

The optimization objective can be expressed as:

$$L = L_0 + S_P \triangle P + S_Q \triangle Q \tag{16}$$

where L_0 represents the initial L index of the AC system; S_P and S_Q represent the sensitivity; ΔP and ΔQ represent the active power and reactive power support.

The equality constraints can be specifically expressed as:

$$P_{Gi} - P_{Li} = U_i \sum_{j=1}^{N} U_j [G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}] \tag{17}$$

$$Q_{Gi} - Q_{Li} = U_i \sum_{j=1}^{N} U_j [G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}] \tag{18}$$

where P_{Gi} and P_{Li} represent the active power generation and load at node i, respectively; Q_{Gi} and Q_{Li} represent the reactive power generation and load at node i, respectively; G_{ij}, B_{ij}, and θ_{ij} represent the conductance, susceptance, and voltage angle difference between nodes i and j respectively; U_i and U_j represent the voltage magnitudes at nodes i and j.

The inequality constraints can be specifically expressed as:

$$U_{i\min} \le U_i \le U_{i\max} \tag{19}$$

where U_{imin}, U_{imax}, and U_i represent the lower limit, upper limit, and actual voltage at node i, respectively.

$$S_{Gi\min} \le S_{Gi} \le S_{Gi\max} \tag{20}$$

where S_{Gimin}, S_{Gimax}, and S_{Gi} represent the lower limit, upper limit, and actual apparent power of i generator, respectively.

$$0 \le P_0 + \triangle P \le S \tag{21}$$

$$-S \le Q_0 + \triangle Q \le S \tag{22}$$

$$\sqrt{(P_0 + \triangle P)^2 + (Q_0 + \triangle Q)^2} \le S \tag{23}$$

where S represents the transmission power limit of the VSC-HVDC system; P_0 and Q_0 represent the initial active power and initial reactive power respectively.

C. The Solution of the Optimization Model

The essence of the optimization model is a multi-variable nonlinear optimization problem. In this paper, a PSO algorithm with a penalty function is employed. The penalty function is used to penalize solutions that violate constraints. The specific process is shown in the Fig. 2.

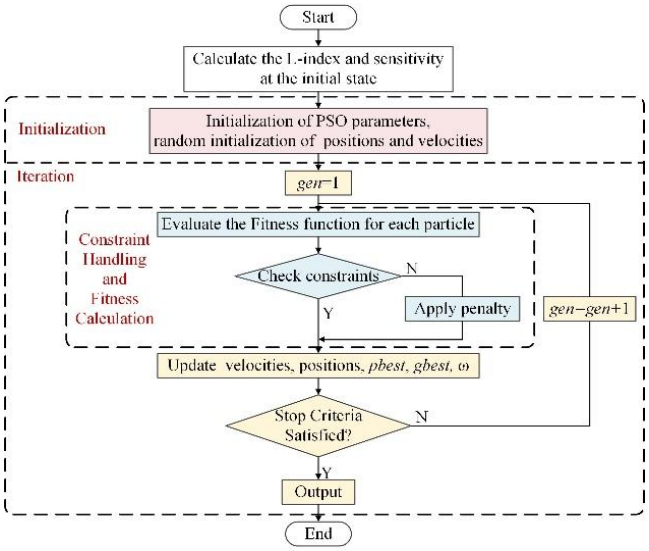


Fig. 2 Optimization Flowchart

Step 1: Calculate the L index and its sensitivity to different variables under the initial operating conditions;

Step 2: Initialize the positions and velocities of the particles;

Step 3: Calculate the fitness of the particles and determine whether they satisfy the constraints. If satisfied, proceed directly to Step 5; otherwise, proceed to Step 4;

Step 4: Assign penalty values to particles that do not satisfy the constraints;

Step 5: Update the velocities, positions, global best, individual best, and inertia weight coefficients of the particles;

Step 6: Determine whether the stopping criterion is met. If satisfied, proceed to Step 7; otherwise, set *gen*=*gen*+1 and return to Step 3;

Step 7: Output the results.

IV. Case Study

A. System Under Study

To verify the effectiveness of the proposed coordinated control strategy the system shown in the figure is taken as an example. In this system, the sending-end system transmits power to the receiving-end 9-node AC network through MMC-HVDC.

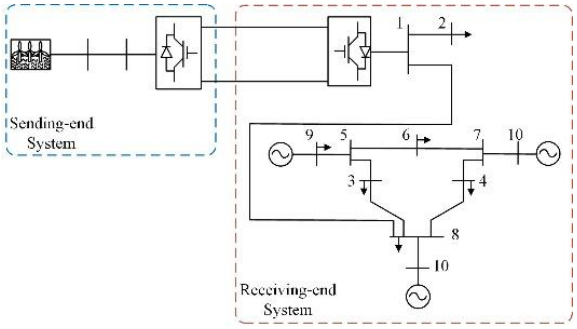


Fig. 3 Simulation System Architecture Diagram

B. Verification of the L Index and Sensitivity Analysis

The initial L index of the system is 0.8810. The sensitivity is calculated using both the perturbation method and the analytical method. The results and errors of the two methods

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are shown in Table I. As can be seen from the table, the error is within 3%, which validates the correctness of the program for solving the sensitivity. The results can be further applied in the coordinated active and reactive power control strategy according to (16).

TABLE I. VERIFICATION OF THE LOAD SENSITIVITY OF THE SYSTEM'S L INDEX

Sensitivity	Perturbation Method	Analytical Method	Error
$\Delta L/\Delta P_1$	-0.27800	-0.28951	2.3147%
$\Delta L/\Delta P_2$	-0.30094	-0.31184	2.2397%
$\Delta L/\Delta P_3$	-0.85076	-0.83840	0.4878%
$\Delta L/\Delta P_4$	-1.01708	-0.99700	0.0479%
$\Delta L/\Delta P_5$	-1.37302	-1.33826	0.8106%
$\Delta L/\Delta P_6$	-1.47072	-1.42981	1.0663%
$\Delta L/\Delta P_7$	-1.37347	-1.33866	0.8193%
$\Delta L/\Delta P_8$	-0.33917	-0.33884	2.0829%
$\Delta L/\Delta P_9$	-1.40039	-1.36591	0.7877%
$\Delta L/\Delta P_{10}$	-1.40077	-1.36618	0.7994%
$\Delta L/\Delta Q_1$	-0.57016	-0.53254	2.6387%
$\Delta L/\Delta Q_2$	-0.56355	-0.52666	2.5816%
$\Delta L/\Delta Q_3$	-0.54299	-0.51700	2.9639%
$\Delta L/\Delta Q_4$	-0.54994	-0.52232	3.2042%
$\Delta L/\Delta Q_5$	-0.15171	-0.15070	1.2434%
$\Delta L/\Delta Q_6$	-0.14587	-0.14474	1.1592%
$\Delta L/\Delta Q_7$	-0.12227	-0.12182	1.6059%
$\Delta L/\Delta Q_8$	-0.56328	-0.53585	2.8296%

C. The Impact of Active and Reactive Power Support on the Static Voltage Stability of AC System

As illustrated above, the active and reactive power support affects the static voltage stability of the AC system directly. To show the impact of power support from VSC-HVDC connected renewable energy on the receiving AC system, the L index of the AC system under different active and reactive power support conditions is calculated, and shown in Fig 4 and 5.

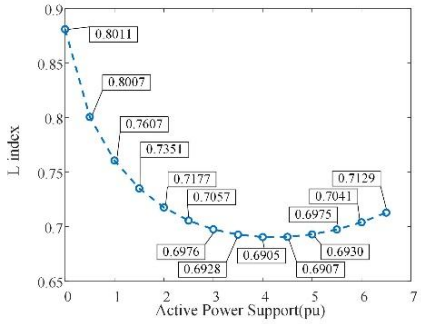


Fig. 4 The Impact of Different Active Power Outputs of the Converter Station on the L Index

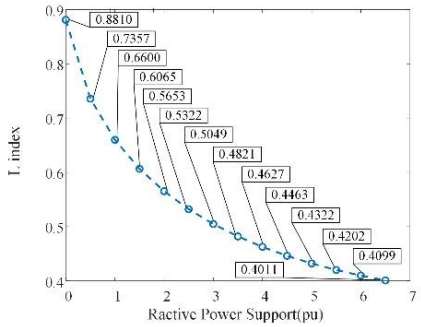


Fig. 5 The Impact of Different Reactive Power Outputs of the Converter Station on the L Index

From the figures, it can be seen that as the active power support provided by the VSC-HVDC system increases, the L index of the AC system first decreases and then increases, but it remains lower than the initial L index. As the reactive power support provided by the VSC-HVDC system increases, the L index of the AC system gradually decreases. This indicates that providing active and reactive power support to the AC system can improve its static voltage stability. Moreover, when the active power support increases from 0 to 4 pu, the L index decreases from 0.8810 to 0.6905. When the reactive power support increases from 0 to 4 pu, the L index decreases from 0.8810 to 0.4627. Compared to active power support, reactive power support is more effective in improving the static voltage stability, which is consistent with the sensitivity results in Table I.

D. Discussions on the Results of Coordinated Active and Reactive Power Control

To verify the effectiveness of the proposed method, this paper compares the following three scenarios:

Scenario 1: The VSC-HVDC system provides only active power support.

Scenario 2: The VSC-HVDC system provides only reactive power support.

Scenario 3: The VSC-HVDC system provides coordinated active and reactive power control.

The initial operating condition of the VSC-HVDC system is P_0=0.8S, Q_0=0. When the VSC-HVDC system provides only active power support, the reactive power remains at the initial state Q_0, with the available active power support ranging from -0.8S to 0.2S. When the system provides only reactive power support, the active power remains at the initial state P_0, with the available reactive power support ranging from -0.6S to 0.6S. When the VSC-HVDC system provides both active and reactive power support, the available active power support ranges from -0.8S to 0.2S, and the available reactive power support ranges from **-S** to **S**. The allowable voltage range at the AC system nodes is 0.9 pu to 1.1 pu.

For the PSO algorithm, the swarm size is set to 500 particles, with a maximum of 200 iterations. Table II shows the L index of the system after applying the power support under the three scenarios.

TABLE II. COMPARISON OF THE L-INDEX IN THREE SCENARIOS

Scenario	Active Power Support	Reactive Power Support	L Index
Scenario 1	0.1984S	0	0.6930
Scenario 2	0	0.3029S	0.5343
Scenario 3	0.1367S	0.2654S	0.5260

From the table, it can be seen that when the VSC-HVDC system provides coordinated control, the L index of the AC system is the smallest, and the static voltage stability is the highest. When the VSC-HVDC system provides only active power support, the L index is significantly larger than in the other two cases. As shown in Fig 4, when the VSC-HVDC system provides only active power support, the minimum L index is approximately 0.6905, which never reaches 0.5206.

When the VSC-HVDC system provides only reactive power support, the L index is slightly larger than that in Scenario 3,

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indicating a slightly worse effect on improving the static voltage stability compared to coordinated control. If the L index in Scenario 2 is also required to reach 0.5260, a reactive power support of 37.53% S would be needed, at which point the maximum node voltage in the system would be 1.1243pu., leading to voltage over-limit issues. Therefore, under actual operating requirements, coordinated active and reactive power control can provide the best performance in improving the static voltage stability.

V. CONCLUSION

This paper proposes a coordinated active and reactive power control strategy for VSC-HVDC connected renewable energy to improve system static voltage stability represented by the L index. The sensitivity of L index to different variables are theoretically explained and validated through simulations, presenting a calculation error that is less than 3%. Compared with the methods providing only active power support, the proposed coordinated control strategy can improve the L index by almost 25%. Compared with the methods providing only reactive power support, the proposed strategy can not only better enhance the L index but also ensure that the system's voltage is within the safety limit. Comprehensive case studies have demonstrated the effectiveness of the proposed control strategy in improving system voltage stability. In the future, the impact of the coordinated control strategy on other voltage stability indexes will be investigated.

REFERENCES

[1] R Yan, TK Saha, F Bai, H Gu. "The anatomy of the 2016 south Australia blackout: A catastrophic event in a high renewable network". IEEE Transactions on Power Systems, vol. 33, no. 5, pp. 5374-5388,2018.

[2] S Imai, D Novosel, D Karlsson, A Apostolov. "Unexpected consequences: global blackout experiences and preventive solutions". IEEE Power and Energy Magazine, vol. 21, no. 3, pp. 16-29, 2023.

[3] W. Wang, W. Lin, G. He, W. Shi, and S. Feng, "Enlightenment of 2021 Texas blackout to the renewable energy development in China", Proceedings of the CSEE, vol. 41, no. 12, pp. 4033-4043, 2021.

[4] RAO Hong, ZHOU Yuebin, LI Weiwei, ZOU Changyue, WANG Zeqing. "Engineering Application and Development Prospects of VSC-HVDC Technology". Automation of Electric Power Systems, vol. 47, no. 1, pp. 1-11, 2023.

[5] T Wang, S Wang, S Ma, J Guo, X Zhou "An extended continuation power flow method for static voltage stability assessment of renewable power generation-penetrated power systems". IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 71, no. 2, pp. 892-896, 2024.

[6] XU Xingdong, WANG Lei, DENG Xiaofan, WANG Cong, SUN Jianchao. "Optimal generators re-dispatch method to meet the enhancement requirements of static voltage stability". Power System Protection and Control, vol. 50, no. 12, pp. 82-89, 2022.

[7] WANG Junheng. "Improving Static Voltage Stability Using Minimum Eigenvalue Sensitivity Method". Zhengzhou University, 2014.

[8] A Urquidez and L Xie. "Singular value sensitivity based optimal control of embedded VSC-HVDC for steady-state voltage stability enhancement". IEEE Transactions on Power Systems, vol. 31, no. 1, pp. 216-225, 2016.

[9] J Zhang, Y Luo, B Wang, C Lu, J Si, J Song. "Deep reinforcement learning for load shedding against short-term voltage instability in large power systems". IEEE Transactions on Neural Networks and Learning Systems, vol. 34, no. 8, pp. 4249-4260, 2023.

[10] LI Shenghu, XIA Weijian, YE Jianqiao, ZHANG Aobo, ZHANG Xiaoyan. "Exchange Relation with Eigenvalues and Threshold Determination of L Index for Power System Static Voltage Stability Analysis". Power System Technology, vol. 47, no. 9, pp. 3828-3837, 2023.

[11] WU Yanning, LUO Yi, LEI Cheng, HUANG Yu, LIANG Yu, ZHOU Shengcun, et al. "Steady-State Voltage Stability Evaluation of Power System Containing Wind Farm Based on Improved PEM and L Index ". Electric Power, vol. 55, no. 9, pp. 193-203, 2022.

[12] LE Jian, LIAO Xiaobing, LI Ben, ZHOU Ziheng, PENG Xuelin. "Global Sensitivity Analysis of Uncertain Static Voltage Stability Based on Extended Affine Model". Transactions of China Electrotechnical Society, vol. 36, no. 13, pp. 2821-2831, 2021.

[13] CHEN Houhe, HUANG Yulei, JIANG Tao, LI Xue, LI Guoqin. "Voltage Stability Analysis and Control of AC/DC Systems with VSC-HVDC". Power System Technology, vol. 41, no. 8, pp. 2429-2438, 2017.

[14] GUO Xiaoying, TANG Junjie, SHU Tong, PENG Sui, XIE Kaigui. "Static Voltage Stability Assessment on Hybrid Alternating Current/Voltage Source Converter-Multiple Terminal Direct Current System Using Improved Modal Analysis". Transactions of China Electrotechnical Society, vol. 36, no. 17, pp. 3741-3752, 2021.

[15] TAN Zukuang. "Research on Access Point and Influence of Large Scale Offshore Wind Farm". Shandong University, 2022.

[16] LIN Yi, TANG Yuchen, OU Jianxian, WU Wei1, BIAN Xiaoyan, ZHAO Jinbin. "Quantitative Analysis of Voltage Support Strength of Offshore Wind Power Grid-Connected System via VSC-HVDC Considering the influence of Sending-side Grid". Power System Technology, vol.48, no. 11, pp. 4524-4533, 2024.

[17] RAO Hong, HUANG Weihuang, GUO Zhifei, ZHOU Yuebin. "Practical Experience of VSC-HVDC Transmission in Large Grid". High Voltage Engineering, vol.48, no. 9, pp. 3347-3355, 2022.

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